INCH POUND

MIL-PRF-49156B
4 March 1999
Superseding
MIL-L-49156A
01 October 1980

PERFORMANCE SPECIFICATION

EYEPIECE ASSEMBLY, FOR NIGHT VISION GOGGLES, AN/PVS-5( )

This specification is approved by the U.S. Army Communications-Electronics Command (CECOM), Department of the Army, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Eyepiece Assembly, part of the Night Vision Goggles, AN/PVS-5( ) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendation, additions, and deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703-0523, by using the Standardization Document Improvement Proposal (DD Form 1426) or by letter.

AMSC N/A FSC 5855

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:52Z
Check the source to verify that this is the current version before use.

2.1.1 Reference documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2.b).

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-C-804193	-	Eyepiece Assembly and all associated drawings
SM-D-804194	-	Adapter, Eyepiece
SM-D-804195	-	Cell Assembly-Eyepiece

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Construction. The Eyepiece Assembly, referred to hereinafter as the eyepiece assembly, shall be constructed to comply with the performance requirements in the contract, drawing SM-C-804193 as applicable, and as specified herein. The Eyepiece Cell Assembly, which is part of the eyepiece assembly and is referred to hereinafter as the cell assembly, shall be in accordance with drawing SM-D-804195-1 (glass) as applicable and as specified herein. The Eyepiece Adapter, which is part of the eyepiece assembly and is referred to hereinafter as the adapter, shall be in accordance with drawing SM-D-804194 as applicable and as specified herein.

3.2 First article test inspection. When specified in the contract or purchase order, the contractor shall furnish eyepiece assemblies for first article test (FAT) inspection and approval(see 6.2.c).

3.3 Parts and materials. The contractor shall select parts and materials fully capable of meeting all operational and environmental requirements specified herein. The materials specified on the applicable drawings are recommended, but not mandatory. Verification that the assembly meets the overall performance requirements shall be the governing acceptance standard.

3.4 Performance characteristics.

3.4.1 Equivalent focal length. The equivalent focal length of the eyepiece assembly shall be 27.0 millimeters (mm) ± 0.3.

3.4.2 Flange focal distance. The flange focal distance of the cell assembly shall be 0.410 inch ± 0.020 where the measurement is taken from the side of the 1.5490 inch cell assembly diameter closest to the focal plane (see SM-D-804195).

3.4.3 Concentricity.

3.4.3.1 Cell assembly concentricity. The concentricity of the optical axis and the mechanical axis associated with the 1.243 inch diameter shall be as specified on cell assembly drawing SM-D-804195.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:52Z
Check the source to verify that this is the current version before use.

3.4.3.2 Adapter concentricity. The concentricity of the mechanical axis associated with the 1.551 inch diameter and the mechanical axis associated with the 1.244 inch diameter shall be as specified on adapter drawing SM-D-804194.

3.4.4 Linear distortion. The linear distortion of the eyepiece assembly at the edge of the 18 mm format shall be between 7.0 percent and 9.5 percent pincushion distortion.

3.4.5 Field of view. The field of view of the eyepiece assembly shall be $40^{o} \pm 2$ for the 18 mm format.

3.4.6 Exit pupil versus eye relief. The exit pupil diameter of the eyepiece assembly shall be no less than 9.8 mm at an eye relief of 15.0 mm ± 0.2 for a zero diopter setting.

3.4.7 Field flatness/field tilt. The field flatness/field tilt shall deviate no more than ± 1/3 diopter over an 18 mm format with a 40 mm concave radius of curvature at the zero diopter image plane of the eyepiece assembly.

3.4.8 Transmission. The transmission of the eyepiece assembly shall not be less than 78 percent over its full aperture for the spectral output of phosphor type number 10-52 (RCA F2126 or equivalent) (see 3.9.5).

3.4.9 Modulation transfer function (MTF). The MTF values of the eyepiece assembly shall not be less than those specified in Table I when tested using an S-11 photomultiplier tube and a green filter (Wratten No. 58 or equivalent over a 2856 K ± 200 tungsten light source) to simulate P-20 illumination in the test setup.

Table I. Modulation transfer function.

Orientation & Pupil Location[1]	Spatial Frequency[2]			
	10 lp/mm	20 lp/mm	30 lp/mm	40 lp/mm
on-axis & centered	92%	82%	66%	43%
on-axis & 1 mm decentered	87%	78%	60%	40%
on-axis & 2 mm decentered	81%	62%	42%	23%
on-axis & 4 mm decentered	50%	32%	23%	13%
on-axis & 5 mm decentered	28%	17%	12%	6%
14° off-axis radial & centered	90%	78%	60%	39%
14° off-axis tangential & centered	90%	78%	60%	39%

[1]incorporate a 5 mm stop at a 15 mm eye relief
[2]in units of line pairs per millimeter (lp/mm)

3.4.10 Weight. The weight of the cell assembly shall not exceed 65 grams.

3.5 Environmental characteristics. For the purpose of post-environmental testing, "perform properly" shall be defined as the ability to successfully complete, in order, the modulation transfer function (3.4.9), transmission (3.4.8), and flange focal distance (3.4.2) tests.

3.5.1 Vibration. The eyepiece assembly shall not be damaged (see 3.9.1) by vibration in the frequency range of 10 hertz (Hz) to 55 Hz with a double amplitude of 0.050 inch ± 0.005.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:52Z
Check the source to verify that this is the current version before use.

3.5.2 Temperature extremes. The eyepiece assembly shall not be damaged (see 3.9.1) by storage in any temperature from +23°C to +65°C or from +23°C to -51°C. The eyepiece assembly shall not be damaged (see 3.9.1) after being subjected to repetitive temperature changes from +23°C to +65°C in a 5 minute period and between +23°C to -51°C in a 5 minute period. Soak time at temperature extremes shall be sufficient for the entire eyepiece assembly to attain stabilization. Time for temperature stabilization is defined as the amount of time required for the most internal part of the unit under test to achieve the specified temperature.

3.5.3 Humidity. The eyepiece assembly shall not be damaged (see 3.9.1) when subjected to an atmosphere with a relative humidity of 95 percent ± 5 over a temperature range of 20°C to 65°C.

3.5.4 Immersion. The eyepiece assembly shall not be damaged (see 3.9.1) after being immersed in fresh water to a depth of not less than 3 feet for a period of not less than 30 minutes. The eyepiece assembly shall be preconditioned by exposure to a temperature of 45°C ± 3 C° for a period of 2 hours prior to being immersed in the fresh water at a temperature of 18°C ± 5 C°.

3.6 Treatment and painting. Treatment and painting shall be anti-reflective, shall be rust resistant and as specified on the applicable drawings unless otherwise specified in the contract or purchase order.

3.7 Marking. The eyepiece assembly shall be marked in accordance with applicable drawings unless otherwise specified in the contract or purchase order.

3.8 Workmanship. Standards of workmanship shall be such that the assemblies will meet all requirements of this specification and any reference specifications or drawings.

3.9 Technical interpretations. The following technical interpretations are, when referenced in sections 3 and 4, mandatory for this specification.

3.9.1 Damage. Damage is defined as: condensation, fogging, or moisture on any internal surface; failure of any finish; dents or breakage; failure of any hardware, connection, interface, or component.

3.9.2 Ambient (room) temperature. Ambient (room) temperature is defined as 23°C ± 10 C°.

3.9.3 Environmental temperature. Environmental chamber temperatures shall be controlled to within ±5C° of the specified temperature unless otherwise specified.

3.9.4 Optical cell. The optical cell is defined as the cylindrical part of the cell assembly which is in direct contact with the lens elements and spacers, holds the lens elements and spacers in their fixed relative positions, and has no external pins, sleeves, flanges, retainers, or other surfaces or fittings attached to it in any manner.

3.9.5 Phosphor type 10-52. The peak spectral response of this phosphor occurs at a wavelength between 0.510 micrometer (µm) and 0.560 µm. The bandwidth measured at the 10 percent points of the spectral response curve is equal to or less than 0.200 µm. The spectral response at 0.650 µm is equal to or less than 10 percent of the peak spectral response.

3.9.6 g. g is defined as the magnitude of the earth’s mean surface gravitational acceleration with a value of 32.2 feet per second per second.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:52Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of inspection. Inspection shall be classified as follows:

a. First article test inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article test inspection. Each eyepiece assembly in the first article test (FAT) inspection lot shall be subjected to the inspections of Table II. Samples selected from the lot shall be subjected to the inspections of Table III. Sample size shall be as specified in the contract or purchase order (see 6.2.c). Failure of any inspection shall be cause for rejection of that eyepiece assembly and may be cause for failure of the FAT inspection. Inspections may be performed in any order except that the Modulation Transfer Function shall be performed last.

Table II. Unit first article test inspection.

Inspection	Requirement Paragraph
Equivalent focal length	3.4.1
Flange focal distance	3.4.2
Concentricity	3.4.3
Cell assembly concentricity	3.4.3.1
Adapter concentricity	3.4.3.2
Linear distortion	3.4.4
Field of view	3.4.5
Exit pupil versus eye relief	3.4.6
Field flatness/field tilt	3.4.7
Transmission	3.4.8
Modulation transfer function	3.4.9

Table III. Sample first article test inspection.

Inspection	Requirement Paragraph
Weight	3.4.10
Vibration	3.5.1
Temperature extremes	3.5.2
Temperature shock	3.5.3
Immersion	3.5.4
Modulation transfer function	3.4.9

4.2.1 Disposition of FAT inspection samples. After successful completion of the FAT, the FAT units shall be refurbished and delivered in accordance with the contract.

4.3 Conformance inspection. Requirements for conformance testing shall be as specified in the contract or purchase order (see 6.2.e). Unless otherwise specified, all tests shall be performed at ambient temperature (see 3.9.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:52Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or purchase order (see 6.2.g). When actual packaging of material is to be performed by DOD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The eyepiece assembly is part of the AN/PVS-5( ) Night Vision Goggles, used for general purpose viewing during periods of darkness.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.0).

c. Statement identifying that FAT testing is or is not required and, if so, the sampling requirements for FAT.

d. Quantity and schedule for FAT testing and disposition of FAT samples.

e. Statement identifying that the contractor must submit his proposal for conformance inspection containing the specific requirements to be verified and the sampling plan that will be used to provide the desired confidence level.

f. Necessary actions by the contractor in the event of an inspection lot failure.

g. Levels of preservation and packaging (see 5.0).

6.3 Definitions. See 3.9.

6.4 Subject/term/keyword listing.

Eyepiece assembly
Eyepiece cell assembly
Eyepiece adapter

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensive changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:52Z
Check the source to verify that this is the current version before use.

Custodian:
Army - CR
Air Force - 11

Review Activity:
Army – AT
Air Force – 99

Preparing Activity:
Army - CR

Project 5855-0196

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-49156B	**2. DOCUMENT DATE (YYMMDD)** 990304

3. DOCUMENT TITLE
Eyepiece Assembly, for Night Vision Goggles, AN/PVS-5()

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME US Army Communications-Electronics Command	b. TELEPHONE (Include Area Code) (1) Commercial (732) 532-9129 (2) AUTOVON 992-9129
c. ADDRESS (Include Zip Code) ATTN: AMSEL-LC-LEO-E-EP Fort Monmouth, NJ 07703-5023	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T19:52Z
Check the source to verify that this is the current version before use.